

AI Smart Parking System - Technologies & Components

Frontend: React.js (User Interface)

Backend: Python + FastAPI (API & Business Logic)

AI Model: YOLOv8 (Vehicle and Number Plate Detection)

OCR: EasyOCR or PaddleOCR (Reads License Plate Text)

Camera: USB/Web Camera or IP Camera (Captures Vehicle Images)

Database: MySQL or PostgreSQL (Stores Vehicle & Parking Data)

Image Processing: OpenCV (Frame Processing & Enhancement)

Communication: REST API between React and FastAPI

Parking Management: Slot Availability, Entry, Exit, and Allocation Logic

Optional Features: JWT Authentication, Admin Dashboard, Notifications, Payment Integration, Analytics.

System Workflow

Camera → YOLOv8 Detection → Number Plate Detection → OCR → Database Verification → Parking Slot Assignment → Entry/Exit Update → Slot Release.